Task-18 :- Technical Documentation(Nginx and ALB)

1. Introduction

1.1 Purpose

The purpose of this document is to detail the technical implementation of a hybrid web hosting architecture on AWS. This project demonstrates the deployment of multiple websites using two distinct methods: Host-based routing via Nginx Server Blocks and High-Availability distribution via an AWS Application Load Balancer (ALB).

1.2 Scope

This documentation covers:

- Provisioning and configuring Nginx on EC2.
- Implementing Nginx Virtual Hosting (Server Blocks) for Site 1 and Site 2.
- Deploying a Load Balanced architecture for Site 3 and Site 4.
- Securing traffic using Certbot (Let's Encrypt) and AWS Certificate Manager (ACM).
- DNS Management and Routing using Route 53.
- Automation of Flask application services using Systemd and Gunicorn.

2. Overview of DevOps Project

2.1 Definition

This project implements automation, scalability, and high availability by hosting four independent websites. It showcases the ability to manage traffic at both the OS level (Nginx) and the Cloud Infrastructure level (ALB).

2.2 Key Principles (AWS Well-Architected Framework)

- **Security:** SSL/TLS termination at both instance and load balancer levels.
- **Reliability:** Multi-node distribution for Site 3 and Site 4 across Availability Zones.
- **Performance Efficiency:** Nginx reverse proxying and Gunicorn worker management.
- **Cost Optimization:** t3.micro/t2.micro instances (Free Tier eligible).
- **Operational Excellence:** Use of Systemd for process persistence.

2.3 Goals and Objectives

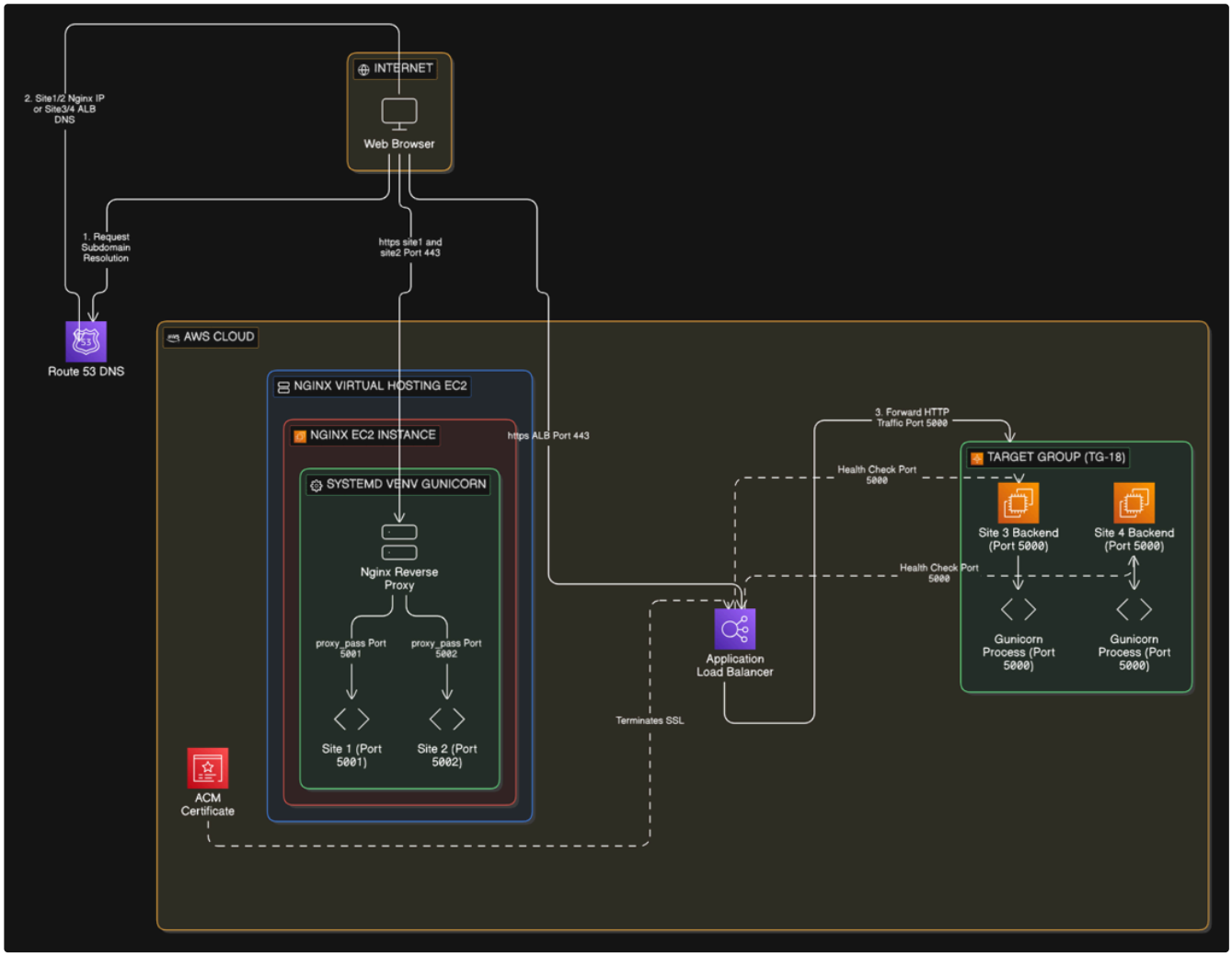
- Host Site 1 and Site 2 on a single EC2 via Nginx port-multiplexing.
- Configure an Application Load Balancer for Site 3 and Site 4.

- Enable HTTPS for all websites (Site 1-4).
- Map all subdomains to `vraj.pushkar.cloud` via Route 53.
- Ensure high availability through ALB health checks.

2.4 Constraints and Assumptions

- AWS Free Tier resources.
- Python Flask application.
- Access to `vraj.pushkar.cloud` Hosted Zone.
- Ports 80, 443, and 5000 open.

3. Architecture and Infrastructure



3.1 Infrastructure Overview

The architecture consists of:

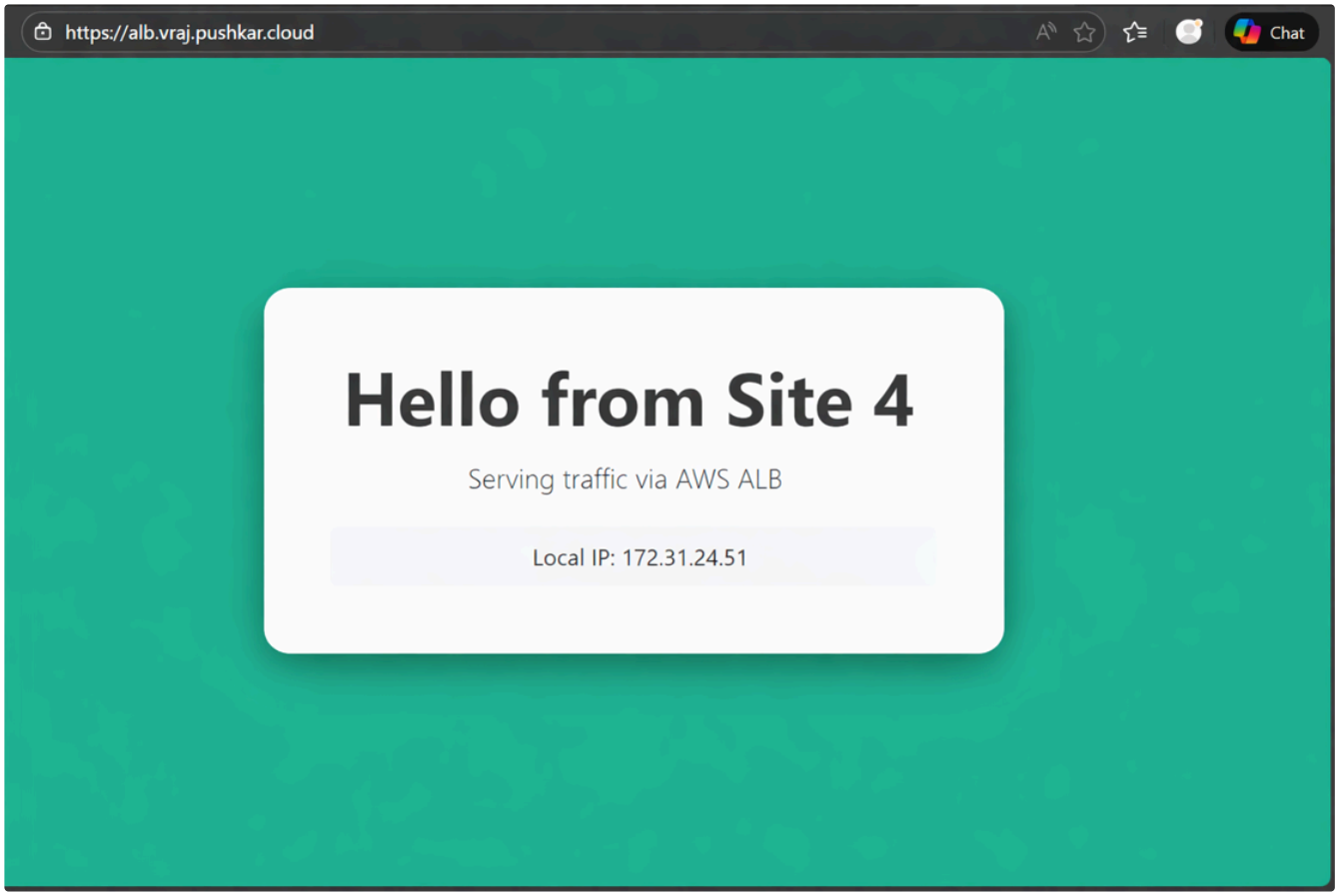
- 1 Nginx Reverse Proxy Instance (Site 1 & 2).

- 2 Backend EC2 Instances (Site 3 & 4).
- Application Load Balancer (ALB).
- Target Group (TG-18).
- AWS Certificate Manager (ACM) & Route 53.

3.2 Environments

Application accessible via:

- `https://site1.vraj.pushkar.cloud`
- `https://site2.vraj.pushkar.cloud`
- `https://alb.vraj.pushkar.cloud` (Routing Site 3 & 4)



3.3 Application Architecture & User Request Flow

- **Direct:** User → Route 53 → Nginx (Port 80/443) → Gunicorn (Port 5001/5002)
- **Load Balanced:** User → Route 53 → ALB (Port 80/443) → Target Group → EC2 (Port 5000)

3.4 Tools and Technologies

- AWS EC2, ALB, Route 53, ACM.
- Nginx, Gunicorn, Flask.
- Certbot (Let's Encrypt).

4. Development Practices

4.1 Version Control

Code is maintained via GitHub repository (pushkar-iamops/hello-world-flask).

4.2 Continuous Integration

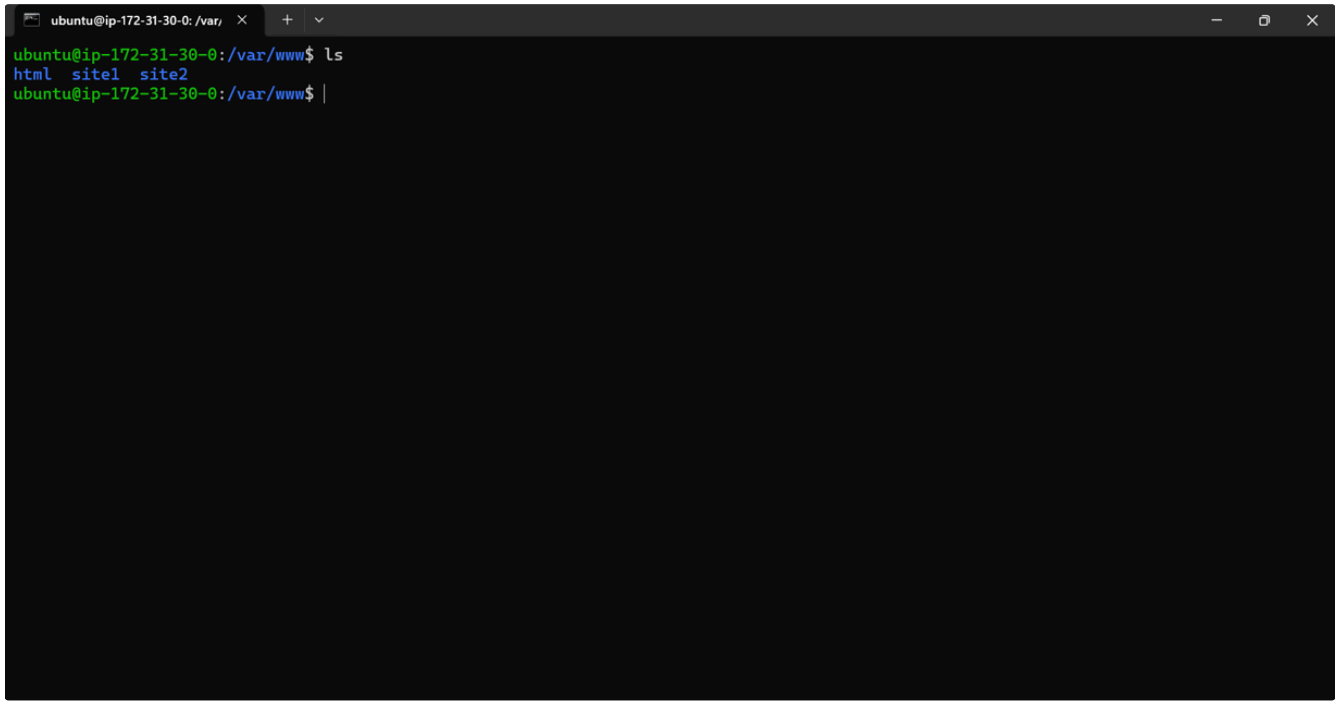
Manual deployment verified via CLI; service persistence managed by Systemd.

5. Deployment and Release Management

5.1 Deployment Strategy

Steps followed:

1. Deploy Flask sites in `/var/www/` directories.
2. Configure Gunicorn for port-specific listening (5000, 5001, 5002).
3. Set up Nginx server blocks for hostname-based routing.
4. Integrate ALB with backend instances and ACM SSL.



Both websites running on the first instance

```
ubuntu@ip-172-31-30-0: /var/
ubuntu@ip-172-31-30-0:/var/www/site2$ sudo systemctl restart site1 site2
ubuntu@ip-172-31-30-0:/var/www/site2$ sudo systemctl status site1 site2
● site1.service - Gunicorn instance to serve Site 1
     Loaded: loaded (/etc/systemd/system/site1.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-03-17 11:17:08 UTC; 6s ago
   Main PID: 4893 (gunicorn)
      Tasks: 5 (limit: 1008)
     Memory: 57.0M (peak: 57.1M)
        CPU: 593ms
     CGroup: /system.slice/site1.service
             ├─4893 /var/www/site1/venv/bin/python3 /var/www/site1/venv/bin/gunicorn --workers 3 --bind 127.0.0.1:5001 hello:app
             ├─4896 /var/www/site1/venv/bin/python3 /var/www/site1/venv/bin/gunicorn --workers 3 --bind 127.0.0.1:5001 hello:app
             ├─4899 /var/www/site1/venv/bin/python3 /var/www/site1/venv/bin/gunicorn --workers 3 --bind 127.0.0.1:5001 hello:app
             └─4901 /var/www/site1/venv/bin/python3 /var/www/site1/venv/bin/gunicorn --workers 3 --bind 127.0.0.1:5001 hello:app

Mar 17 11:17:08 ip-172-31-30-0 systemd[1]: Started site1.service - Gunicorn instance to serve Site 1.
Mar 17 11:17:08 ip-172-31-30-0 gunicorn[4893]: [2026-03-17 11:17:08 +0000] [4893] [INFO] Starting gunicorn 25.1.0
Mar 17 11:17:08 ip-172-31-30-0 gunicorn[4893]: [2026-03-17 11:17:08 +0000] [4893] [INFO] Listening at: http://127.0.0.1:5001 (4893)
Mar 17 11:17:08 ip-172-31-30-0 gunicorn[4893]: [2026-03-17 11:17:08 +0000] [4893] [INFO] Using worker: sync
Mar 17 11:17:08 ip-172-31-30-0 gunicorn[4893]: [2026-03-17 11:17:08 +0000] [4893] [INFO] Control socket listening at /var/www/site1/gunicorn.ctl
Mar 17 11:17:08 ip-172-31-30-0 gunicorn[4896]: [2026-03-17 11:17:08 +0000] [4896] [INFO] Booting worker with pid: 4896
Mar 17 11:17:08 ip-172-31-30-0 gunicorn[4899]: [2026-03-17 11:17:08 +0000] [4899] [INFO] Booting worker with pid: 4899
Mar 17 11:17:08 ip-172-31-30-0 gunicorn[4901]: [2026-03-17 11:17:08 +0000] [4901] [INFO] Booting worker with pid: 4901

● site2.service - Gunicorn instance to serve Site 2
     Loaded: loaded (/etc/systemd/system/site2.service; enabled; preset: enabled)
     Active: active (running) since Tue 2026-03-17 11:17:08 UTC; 6s ago
   Main PID: 4894 (gunicorn)
      Tasks: 5 (limit: 1008)
     Memory: 57.0M (peak: 57.2M)
        CPU: 603ms
     CGroup: /system.slice/site2.service
             ├─4894 /var/www/site2/venv/bin/python3 /var/www/site2/venv/bin/gunicorn --workers 3 --bind 127.0.0.1:5002 hello:app
             ├─4898 /var/www/site2/venv/bin/python3 /var/www/site2/venv/bin/gunicorn --workers 3 --bind 127.0.0.1:5002 hello:app
             ├─4900 /var/www/site2/venv/bin/python3 /var/www/site2/venv/bin/gunicorn --workers 3 --bind 127.0.0.1:5002 hello:app
             └─4902 /var/www/site2/venv/bin/python3 /var/www/site2/venv/bin/gunicorn --workers 3 --bind 127.0.0.1:5002 hello:app
```

5.2 Configuration Management

Gunicorn configured as systemd service for Site 1 and Site 2 to ensure the applications start on boot.

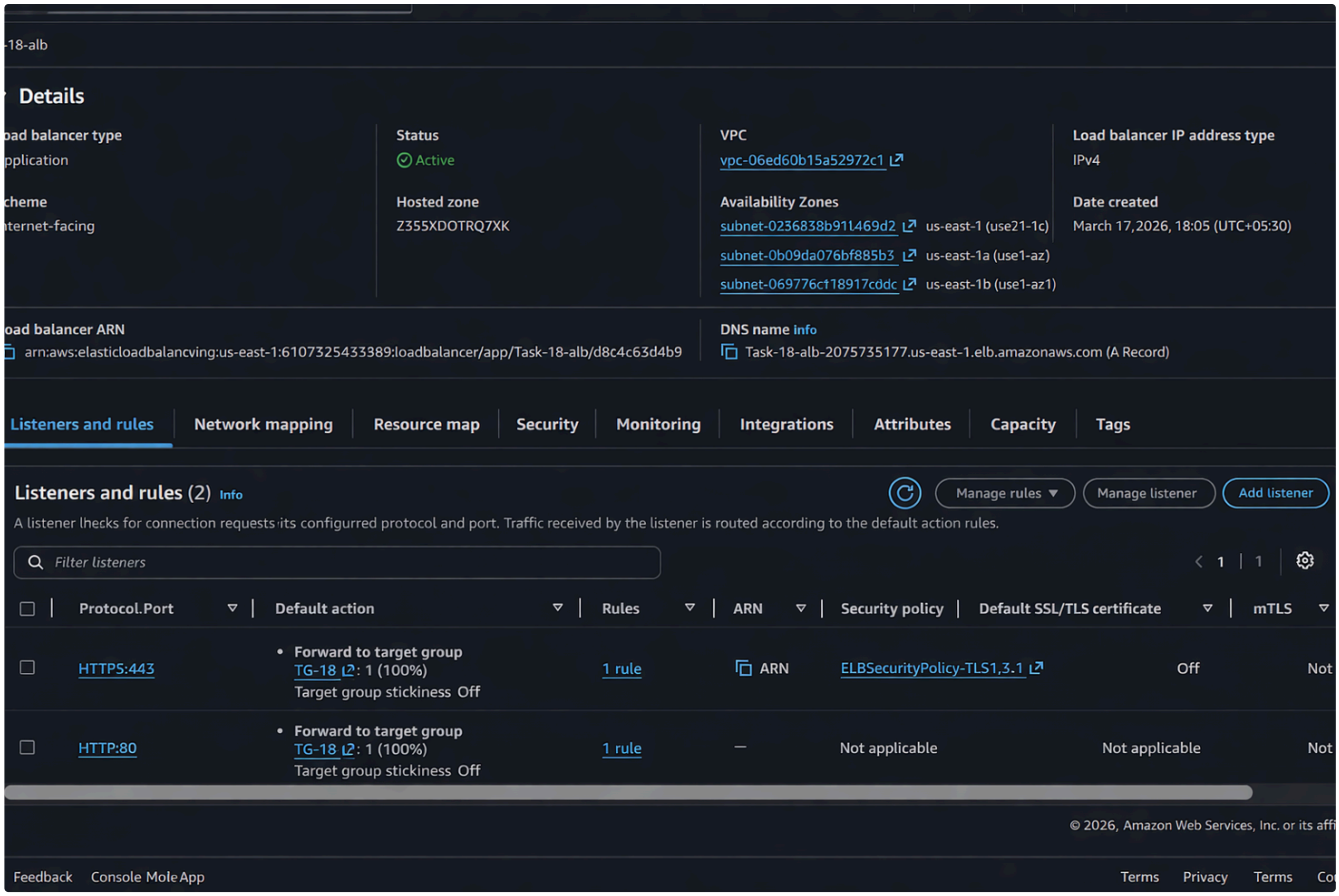
Systemd Service Configuration (/etc/systemd/system/site1.service):

```
Ini, TOML
```

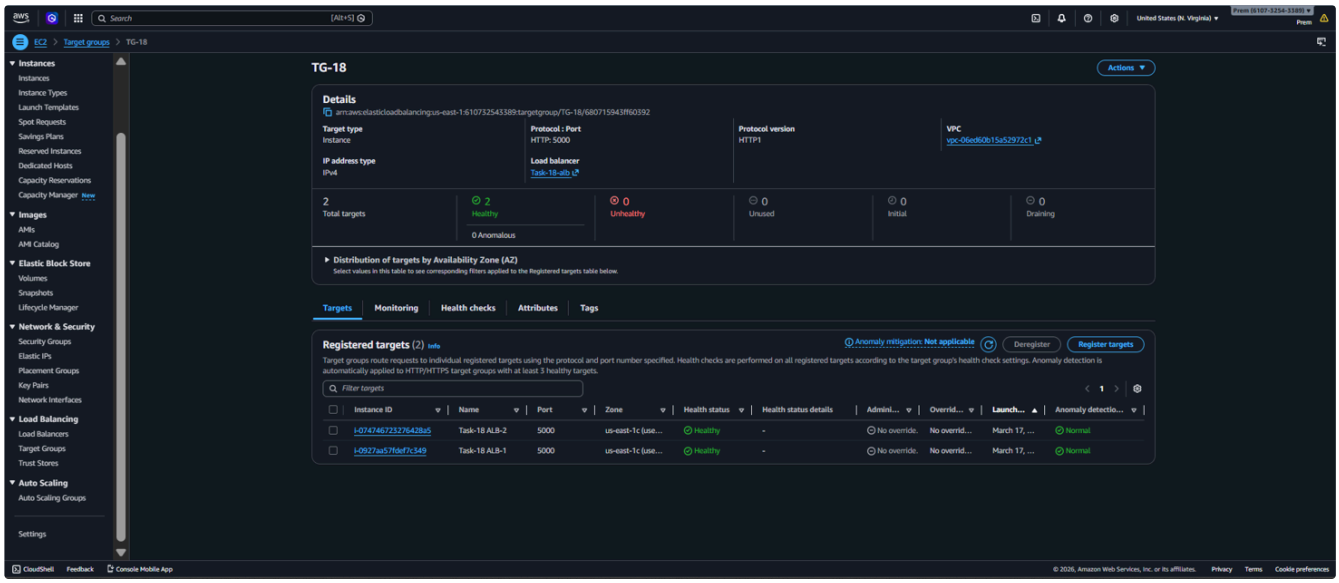
```
[Unit]
Description=Gunicorn instance to serve Site 1
After=network.target
[Service]
User=ubuntu
Group=www-data
WorkingDirectory=/var/www/site1
Environment="PATH=/var/www/site1/venv/bin"
ExecStart=/var/www/site1/venv/bin/gunicorn --workers 3 --bind 127.0.0.1:5001 hello:app
[Install]
WantedBy=multi-user.target

```

5.3 Load Balancer Configuration

- ALB created with HTTP/HTTPS listeners.
- Target group `TG-18` configured on port 5000.
- Health checks enabled on path `/`.

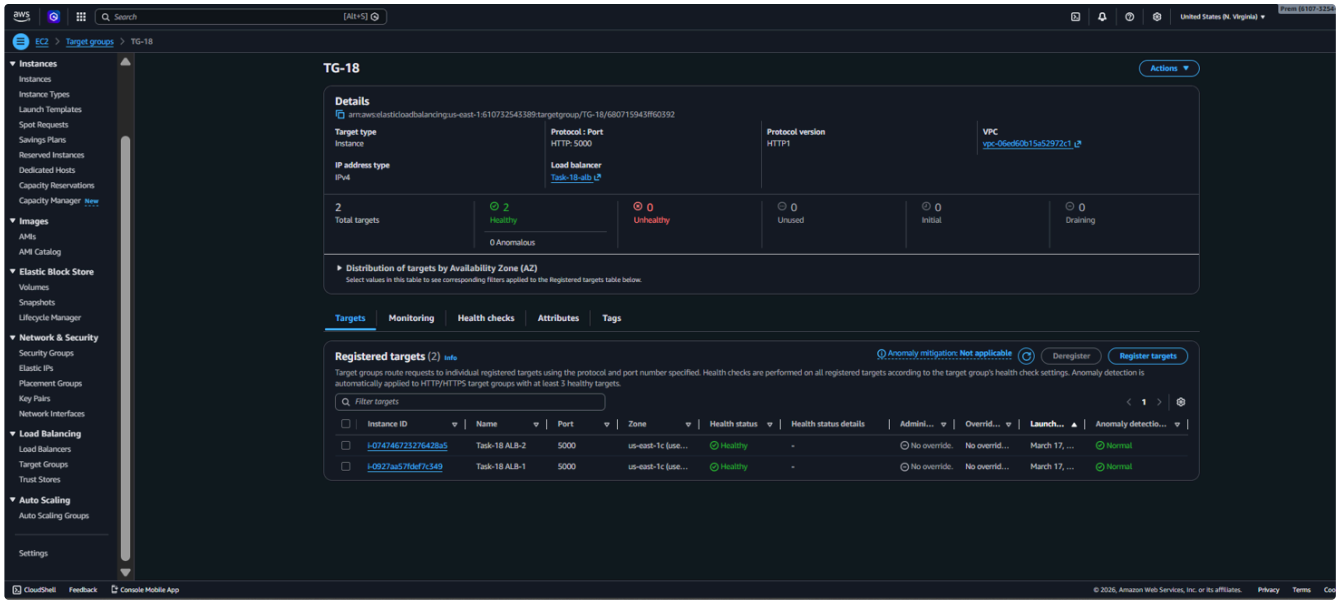


TG created



5.4 Target Group Health

Both Site 3 and Site 4 instances are healthy and receiving traffic from the ALB.



6. SSL and DNS Implementation

6.1 Certbot (Site 1 & 2)

Certbot used to secure Nginx hosts directly on the instance.

Nginx Server Block Configuration (/etc/nginx/sites-available/multiplex):

```
Nginx
```

```
server {
    listen 80;
    server_name site1.vraj.pushkar.cloud;
    location / {
        proxy_pass http://127.0.0.1:5001;
        include proxy_params;
    }
}
server {
    listen 80;
    server_name site2.vraj.pushkar.cloud;
    location / {
        proxy_pass http://127.0.0.1:5002;
        include proxy_params;
    }
}

```

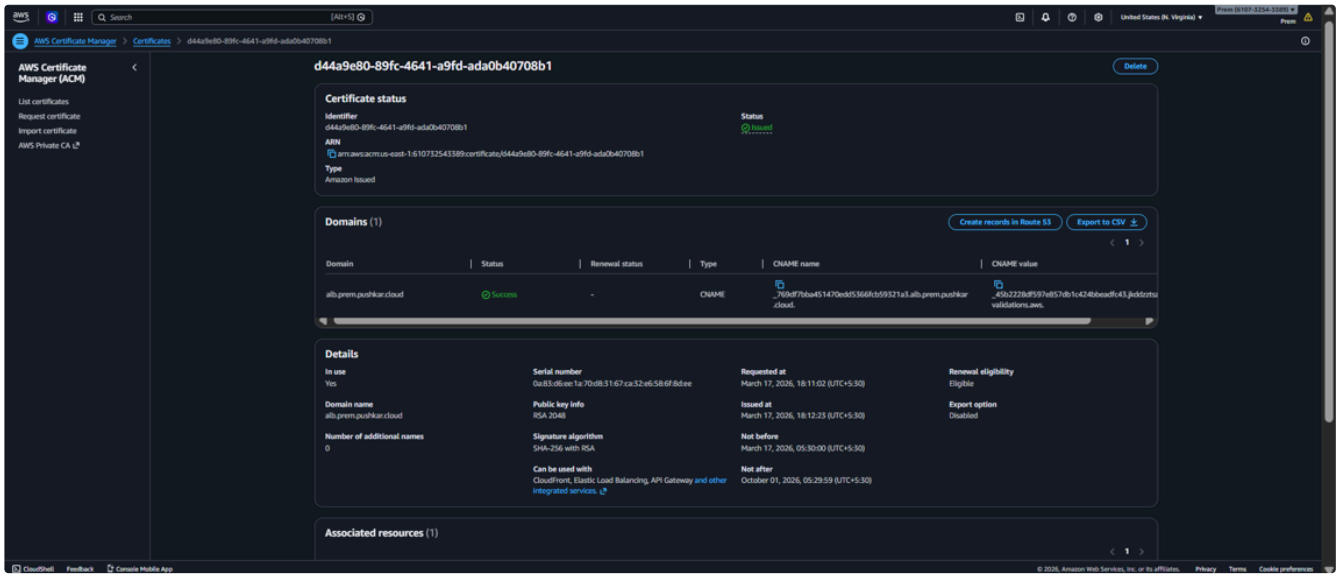
Certbot certificates:-

```
ubuntu@ip-172-31-30-0: /var/
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
Would you be willing, once your first certificate is successfully issued, to
share your email address with the Electronic Frontier Foundation, a founding
partner of the Let's Encrypt project and the non-profit organization that
develops Certbot? We'd like to send you email about our work encrypting the web,
EFF news, campaigns, and ways to support digital freedom.
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
(Y)es/(N)o: n
Account registered.
Requesting a certificate for site1.prem.pushkar.cloud and site2.prem.pushkar.cloud

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/site1.prem.pushkar.cloud/fullchain.pem
Key is saved at:         /etc/letsencrypt/live/site1.prem.pushkar.cloud/privkey.pem
This certificate expires on 2026-06-15.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

Deploying certificate
Successfully deployed certificate for site1.prem.pushkar.cloud to /etc/nginx/sites-enabled/multiplex
Successfully deployed certificate for site2.prem.pushkar.cloud to /etc/nginx/sites-enabled/multiplex
Congratulations! You have successfully enabled HTTPS on https://site1.prem.pushkar.cloud and https://site2.prem.pushkar.cloud

- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
If you like Certbot, please consider supporting our work by:
 * Donating to ISRG / Let's Encrypt:   https://letsencrypt.org/donate
 * Donating to EFF:                    https://eff.org/donate-le
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
ubuntu@ip-172-31-30-0:/var/www$ sudo nginx -t
sudo systemctl restart nginx
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
ubuntu@ip-172-31-30-0:/var/www$
```

6.2 AWS ACM (Site 3 & 4)

ACM Certificate issued for `alb.vraj.pushkar.cloud` and attached to ALB for centralized SSL termination.

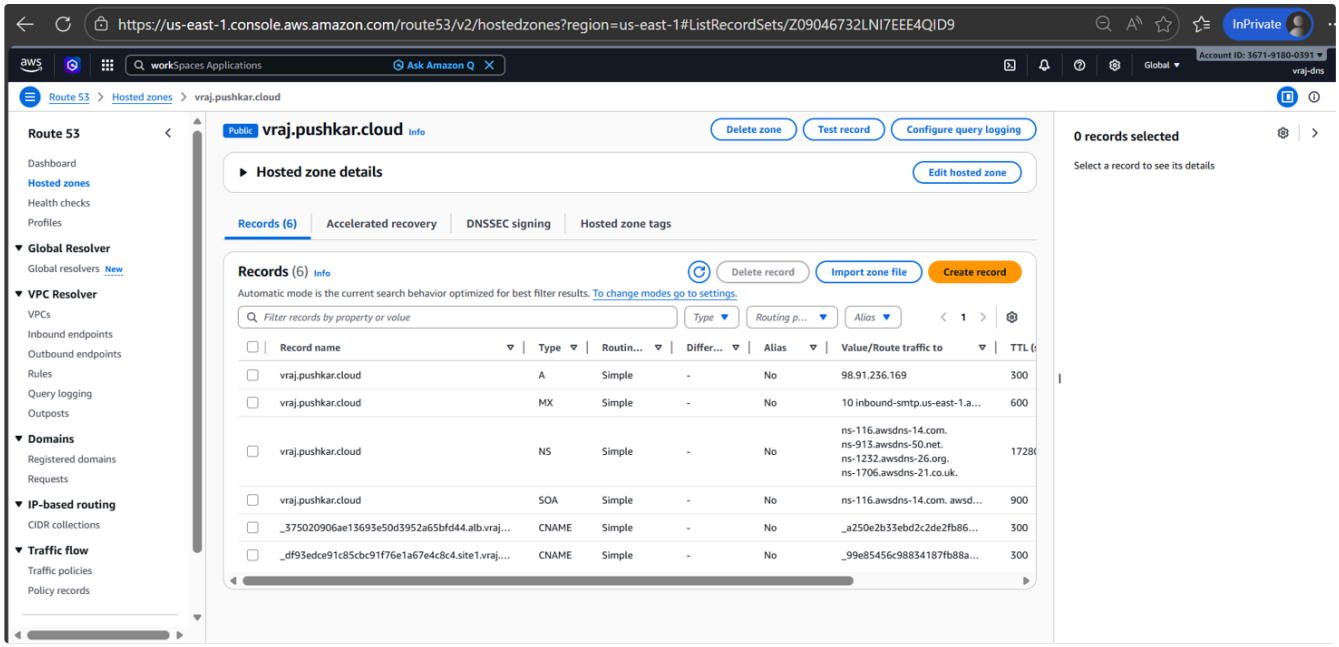
ACM issued



6.3 Route 53 DNS Setup

Records created for Site 1 (A), Site 2 (A), and ALB (CNAME/Alias).

DNS records:-

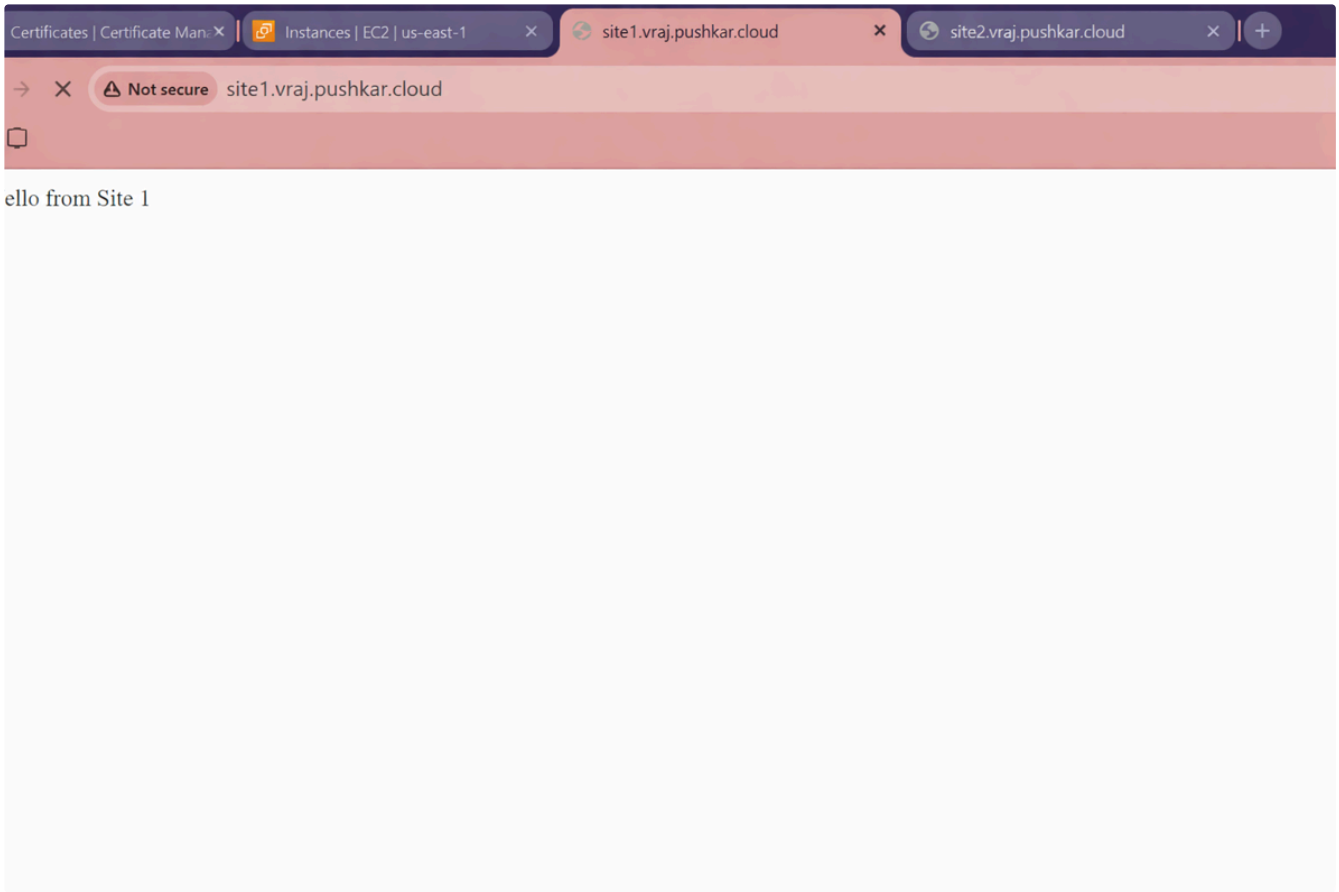


7. Application Validation

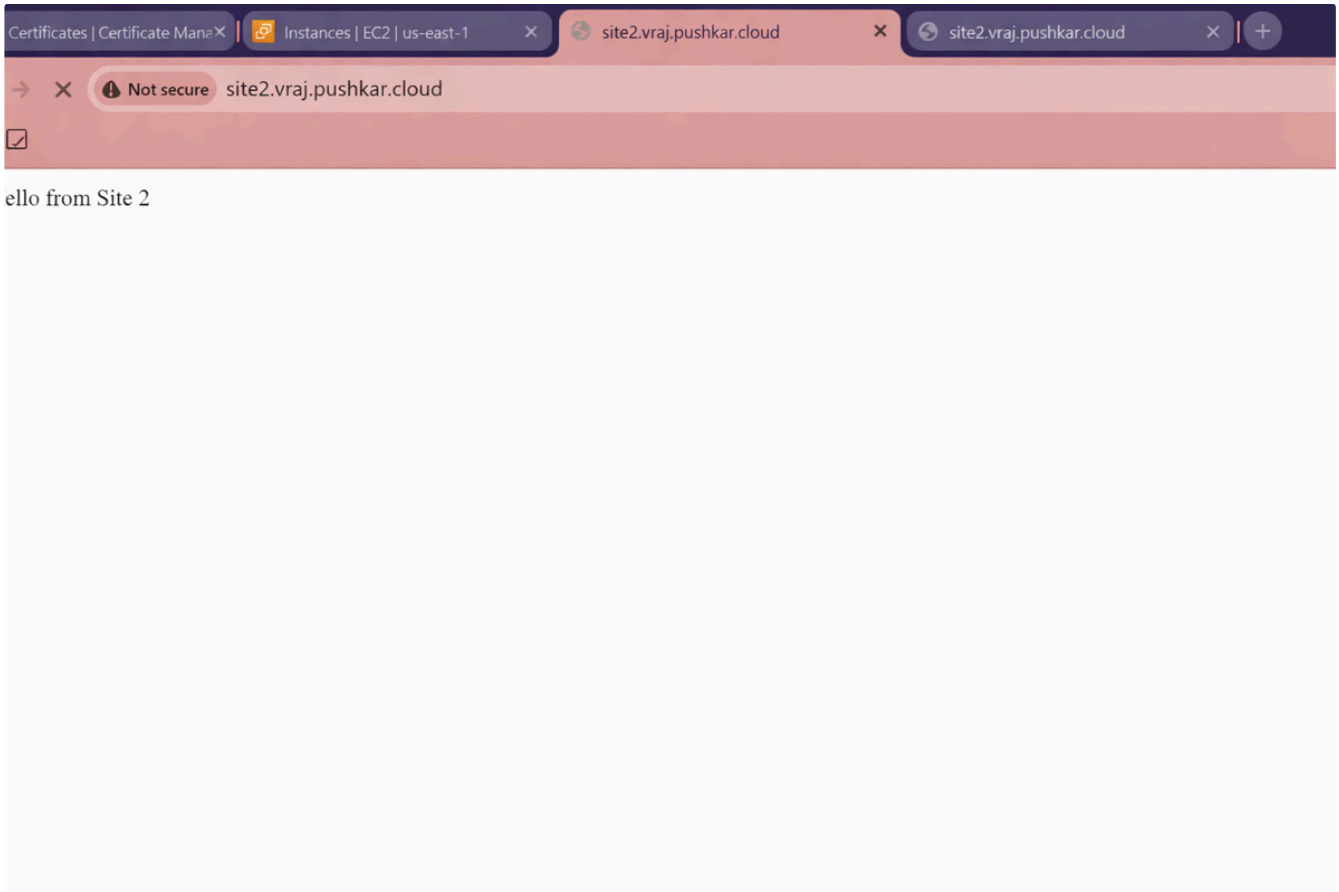
7.1 Flask Application Running

Applications are successfully running on all backend ports (5001, 5002, and 5000).

hello from site1:-



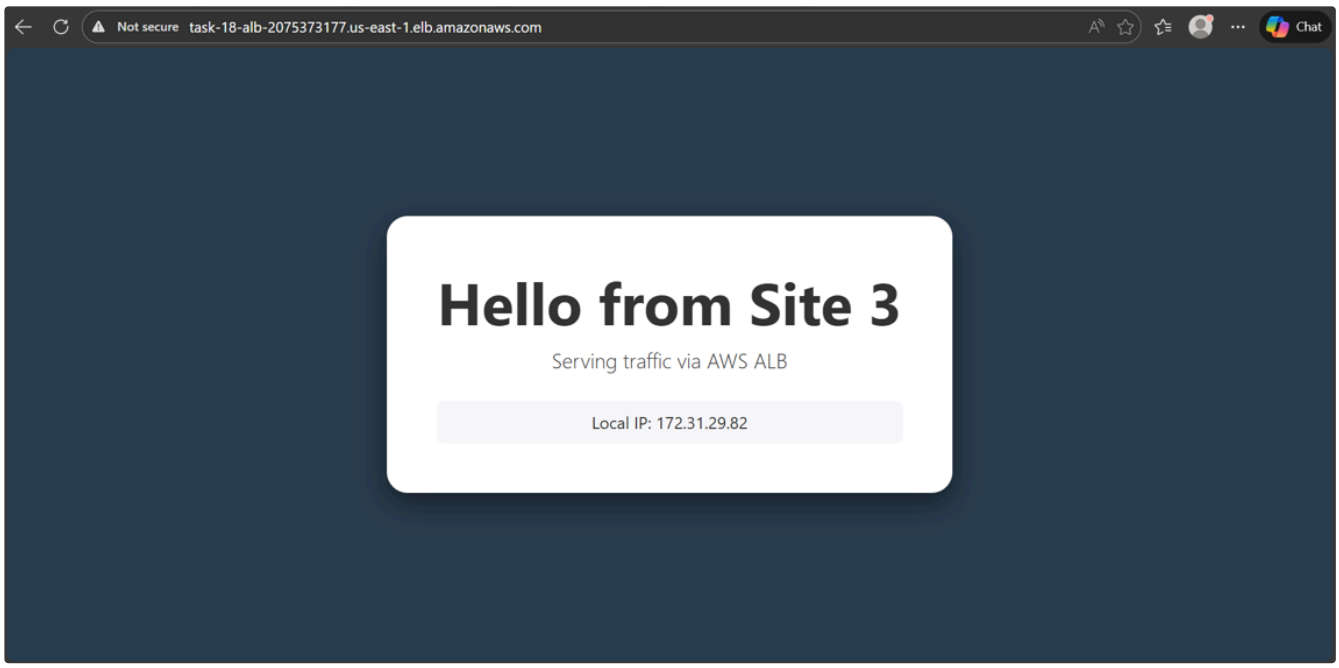
hello from site2 http



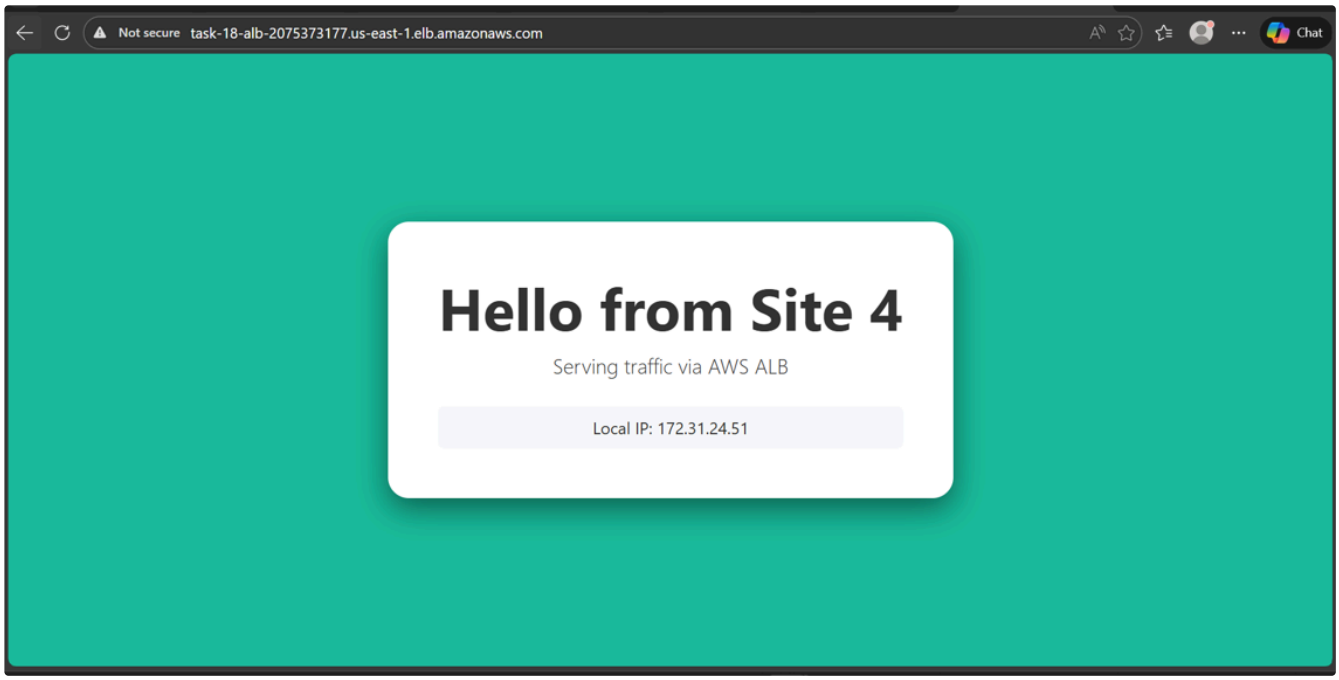
7.2 Application via Load Balancer

Site 3 and Site 4 are accessible and secured through the ALB.

Task-18 alb-site-3



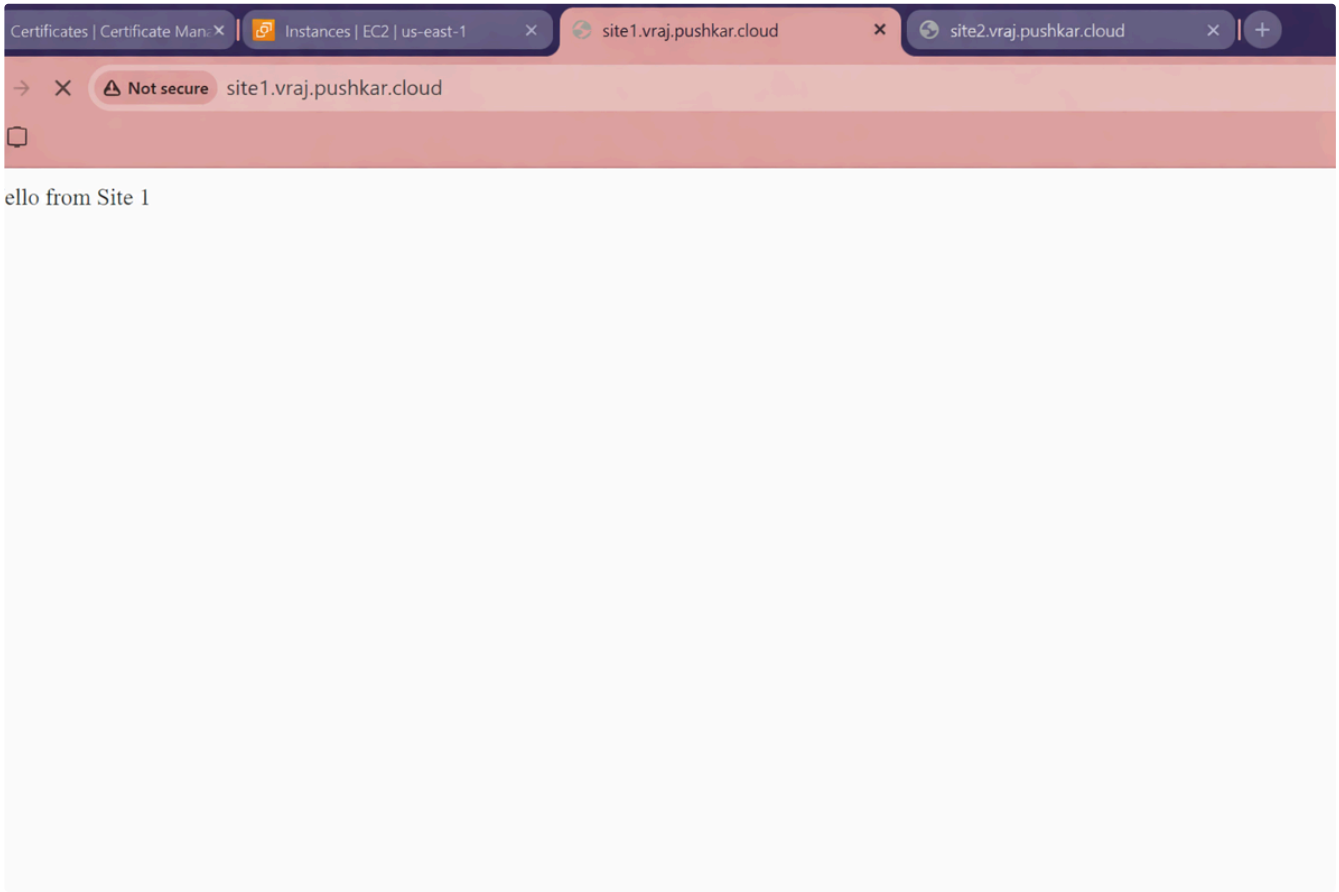
Task-18 alb-site-4



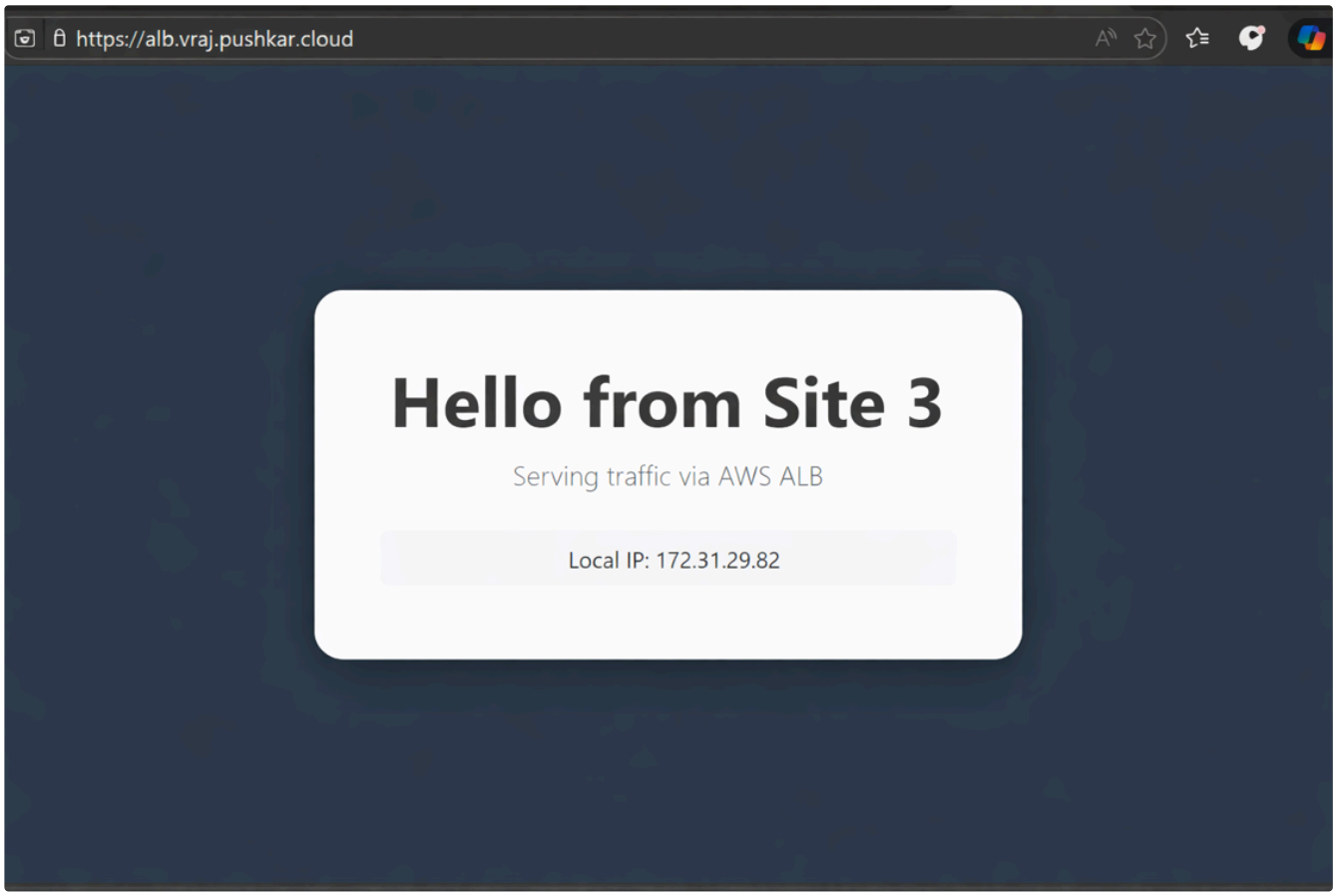
7.3 HTTPS and Content Verification

Verified padlock icons and correct "Hello from Site X" content using Bootstrap templates.

hello from site1 https



alb https created and reply from site4



8. Monitoring and Incident Management

8.1 Monitoring Tools

- `systemctl status site1 site2`
- ALB Target Health Dashboard
- Nginx access and error logs.

8.2 Alerting

Failures detected via:

- Unhealthy target status in EC2 Console.
- Systemd service "failed" state.

9. Cost Optimization

- **Instance type:** t3.micro/t2.micro.
- **Cost:** $0 (Free Tier eligible).

10. Implementation Timeline

- **Nginx VHost Setup:** 1 hour.
- **Backend EC2 Setup:** 1 hour.
- **ALB & SSL Setup:** 45 mins.
- **DNS & Validation:** 15 mins.

11. Continuous Improvement

11.1 Feedback

- Hybrid SSL management (Certbot + ACM) is successful.
- Manual deployment is stable but could be automated via SSM or Jenkins.

11.2 Enhancements

- Implement Auto Scaling Groups (ASG) for Site 3 and 4.
- Centralize all logs in CloudWatch Logs.

12. Conclusion

12.1 Summary

Successfully implemented a multi-site AWS architecture utilizing Nginx Virtual Hosting and Application Load Balancing with end-to-end SSL encryption.

12.2 Next Steps

- Integrate with a CI/CD pipeline.
- Add CloudFront for global content delivery.

13. Customer Acknowledgement

Name:

Title:

Date: